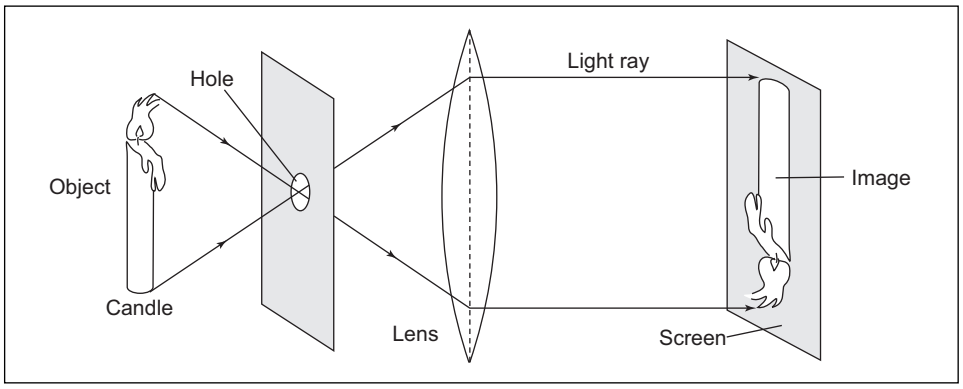
The Optometrist is in

14a

In your science lessons you may have used a candle flame to produce an inverted image when you were studying the eye.



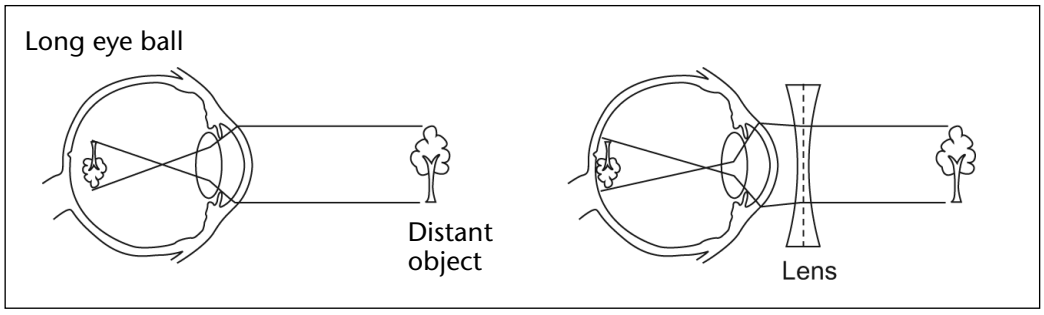
1 Explain what the diagram is representing. Use all the words in the diagram.

The lenses in your eyes perform a similar focus role. The image produced is focused at the back of your eye on the retina. Unfortunately some people have vision problems – a focused and clear image does not fall onto the retina. Long-sightedness and short-sightedness are two very common problems. Below is a table an optometrist developed to help explain to her patients the problems of long-sightedness and short-sightedness and how the problems could be corrected.

Disorder	Problem	Who can be affected	How is it corrected
Long-sightedness	Trouble focusing on close objects. Image focuses behind the retina. Can see distant objects clearly.	People with a short eyeball. People whose lens muscles have weakened or whose lenses have hardened.	Spectacles with converging lenses. These lenses cause light rays to bend inwards.
Short-sightedness	Hard to focus on objects far away. Image is focused in front of retina. Can see close objects clearly.	People who have a long eyeball. People whose lens muscles can no longer produce a thin lens.	Spectacles with diverging lenses. These lenses cause light rays to bend outwards.

2 Use information from the table to explain what is happening in each pair of diagrams.

a

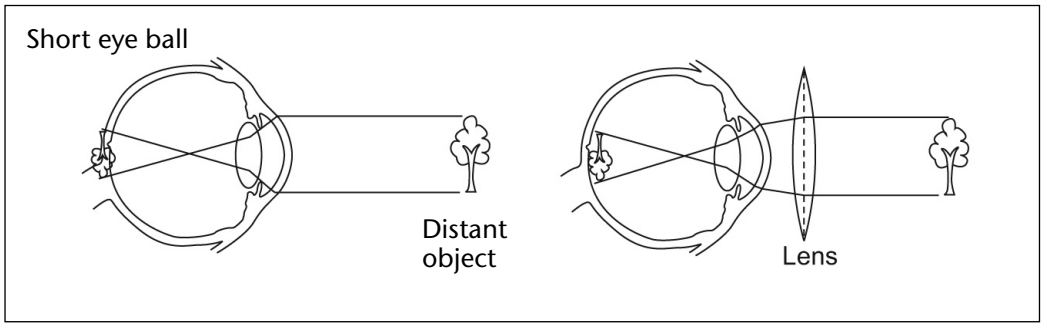


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b



3 Draw a diagram of a healthy eye (show the lens shape and where the image is projected) viewing an object that is:

a far away

b close to the eye.

4 As an optometrist, after examining one of your patients, you discover that she is suffering from long-sightedness. Use your knowledge and expertise in this area to explain to your patient why her vision is blurry and how you will correct it.

5 In your science book, rewrite the information from the table above as a report. Use the report text style. Remember to include a title and an introduction.

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